

Architectural Particular Specification

THERMAL PROTECTION INSULATION
SECTION 6.2

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SECTION 6.2

THERMAL PROTECTION INSULATION

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The work of this Section includes, but is not limited to, the furnishing and installation of the following.
 - 1. Thermal insulation, fixings and required accessories as specified herein and indicated on the Contract Documents.
 - 2. Fire stopping insulation, required accessories and fixings for installation in Section 6.4.
 - 3. Vapour barrier, reinforced tape and other accessories, where specified.
 - 4. Coordinated placement of all insulation and vapour barrier materials and accessories within wall, and floor/ceiling assemblies, and prior to closing of finishes by other trades.
 - 5. Insulation to be installed within nested metal stud joints, headers and corners
 - 6. Raised floor insulation – Thermobreak, or approved equivalent, crosslinked closed cell polyolefin foam with factory applied reinforced aluminum foil

1.02 REFERENCES

- A. British Standards
 - 1. BS5250 : 1989 - Code of Practice for Control of Condensation in Buildings.
- B. American Standards
 - 1. ANSI/UL 263 - Standard For Fire Tests of Building Construction and Materials.
 - 2. ASTM C553 - 92 : Specification for Mineral Fiber Blanket and Thermal Insulation for Commercial and Industrial Applications.
 - 3. ASTM C552: Specification for Cellular Glass Thermal Insulation.
 - 4. ASTM C612-93: Specification for Mineral Fiber Block and Board Thermal Insulation.
 - 5. ASTM C 795-92: Specification for Thermal Insulation for Use in Contact with Austenitic Stainless Steel.
 - 6. ASTM E84-94: Standard Test Method for Surface Burning Characteristics of Building Material.
 - 7. ASTM E 90-90: Standard Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions.
 - 8. ASTM E 96-94: Test Methods for Water Vapour Transmission of Materials.
 - 9. ASTM E492-90 : Standard Test Method for Laboratory Measurement of Impact Sound Transmission Through Floor - Ceiling Assemblies Using The Tapping Machine.

1.03 QUALITY ASSURANCE

- A. Coordinate and carry out the installation of vapour barriers, thermal insulation in sequence with related Work operations to be executed by other trades.

- B. The Contractor shall inspect all voids, cavities, gaps, enclosed spaces and assemblies during the Work construction on every floor to ensure that vapour barrier, thermal insulation are properly installed prior to the enclosing of assemblies or the covering up of all in accessible areas.
- C. All thermal bridging, acoustical and vibration flanking and vapour barrier discontinuity shall be effectively minimized.

1.04 PERFORMANCE REQUIREMENTS

- A. If the ceiling cavities and fan rooms are being used as air plenums, non-combustible material shall be used within all ceiling cavities.
- B. All thermal insulation materials, including facings, such as vapour barriers or breather papers installed within floor-ceiling assemblies, roof-ceiling assemblies, walls, cavities or enclosed spaces (unless more stringent requirements are noted herein) shall have a flame-spread rating not to exceed 25 and a smoke density not to exceed 0 when tested in strict accordance with ASTM E 84-94 and ASTM E119-88.
- C. All insulation shall be certified and containing a label identifying its flame-spread rating, smoke density and approved testing laboratory certificate number.
- D. Thermal insulation shall be inert, durable, rot and vermin proof and shall not be degradable by moisture or water vapour.
- E. The material used and its method of attachment to the supporting component shall be selected to eliminate the risk of bulging, sagging, delaminating or detachment of the insulation.
- F. All insulation forming part of a rated construction, or as required by the statutory authorities, shall be approved by the statutory authorities.

G. Fire and Smoke Performance for polyolefin foam thermos insulation for raised floor

ASTM E84	COMPLIES (NFPA 90A & B) Flame Spread Index: ≤25 Smoke Developed Index: ≤50
BS 476 Parts 6 & 7:	Class 0
EN 13823 (Sheet)	Single Burning Item COMPLIES (EUROCLASS B - s2, d0 RATING)
ISO 11925 Part 2 (Sheet)	Ignitability: COMPLIES (EUROCLASS B - s2, d0 RATING)
FM 4924 (Sheet)	Up to 25mm thickness FM Approved
ISO 5659 Part 2	Smoke Density: COMPLIES (EN 45545-2, R1 HL3 Rating) Smoke Toxicity: COMPLIES (EN 45545-2, R1 HL3 Rating)
AS1530 Part 3	Ignitability Index: 0 Spread of Flame Index: 0 Heat Evolved Index: 0 Smoke Developed Index: 0-1

1.05 ENVIRONMENTAL REQUIREMENTS

- A. All insulation products and materials shall not contain any asbestos or asbestos by-products or known hazardous materials. Insulation fibres shall be entrained and sealed to prevent their release into the Project Building's air return system throughout the lifetime of the building.
- B. The Contractor shall not use materials that generally comprise of mineral fibres, either man-made or naturally occurring, which have a diameter of 3 microns or less and a length of 200 microns or less or which contain any fibres not sealed encapsulated or otherwise stabilised to ensure that fibre migration is prevented.

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

- A. Vapour Barrier, Thermal Insulation Sub-Contractor and the Insulation Manufacturers are deemed to be familiar with climatic conditions prevailing in Hong Kong, and the Project Building's design psychometric requirements, and to be aware of:
 - 1. The high temperature (up to 37oC) and the high relative humidity (up to 95% RH).
 - 2. Psychometric perimeters used by the Consulting Mechanical Engineer in determining vapour barrier, insulation, indoor temperature and relative humidity requirements, heating and cooling loads.

- B. The Thermal Insulation Sub-Contractor (in submitting a Tender providing thermal insulation and technical services, respectively, and executing the Project Work) will assume to warrant that all products, materials and items are suitable, and can be maintained in the various climatic and psychometric conditions encountered without thermal insulation degradation, interstitial condensation within building assemblies, surface condensations, support and growth of moulds, thermal or vapour bridging occurring.

1.07 SUBMITTALS

- A. Submit product and technical performance data, method statement for application, and samples.
- B. Relevant fire test reports shall be submitted for approval before ordering.
- C. Fire resistance test certificates.

1.08 PRE-INSTALLATION CONFERENCE

- A. Record discussions of conference and decisions and agreements reached and furnish copy of record to each party attending and to the Architect for review. Review foreseeable methods and procedures related to vapour barrier, thermal insulation Work, including the following:
 - 1. Tour, inspect and discuss condition of substrate, vapour barrier and insulation materials, and preparatory Work performed by other trades.
 - 2. Review thermal insulation and vapour barrier requirements (Drawings, Specifications and other Contract Documents).
 - 3. Review required submittals, both completed and yet to be completed.
 - 4. Review and finalize Construction Programme of Work related to all Work and verify availability of materials, installer's personnel, equipment and facilities needed to make progress and avoid delays.

1.09 DELIVERY, STORAGE AND HANDLE

- A. Materials shall be delivered in Thermal Insulation Material Manufacturers' dry, undamaged, unopened packages, all clearly labelled with the manufacturer's name, product identification, and certificates.
- B. Materials shall be stored in strict accordance with the Manufacturer's printed instruction, copies of which will be furnished to the Architect. Store under cover, and protect from weather and construction activities.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

(NOT USED)

1.12 WARRANTY

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.
- B. The warranty for the works shall be ten (10) years from the certified date shown on the Certificate of Substantial Completion of the Main Contract issued by Architect. The thermal insulation for raised floor shall be warranted against any defect and loss of performance. The contractor and manufacturer are liable to repair and /or supply of any defective insulation found or arising within the Warranty period (10 years).
- C. The Warranty is conditional upon the following:
 - 1. Insulation should be installed according to the manufacturer's recommendations, including the insulation material and accessories such aluminium foil tape.
 - 2. The thicknesses used should be strictly based on expected operational conditions/data. Thermacalc recommended thickness calculations are based on standard published formulae based on ASHRAE and British Standard recommendations.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURERS/SUPPLIERS

- A. Subject to compliance with the Project Contract Documents, provide thermal insulations, vapour barrier, and required accessories for roofing from one, or more, of the following approved manufacturers/ suppliers, or equivalent approved by the Architect:

Tech Link Construction Engineering Ltd.
Unit 802, 8/F, Laford Centre,
No. 838 Lai Chi Kok Road,
Cheung Sha Wan Kowloon
Tel: (852) 3701 1817 / 9316 8906 Fax: (852) 3701 1818
Email: kenuskwong@techlinkcon.com.hk

1. or approved equivalent.

- B. Subject to compliance with the Project Contract Documents, provide thermal insulations for data hall raised floor system from one, or more, of the following approved manufacturers/ suppliers, or equivalent approved by the Architect:

1. Shun Wah Contracting Co Ltd.
Room 803, 8/F, Block A,
Sea View Estate, No. 2 Watson Road,
North Point, Hong Kong
Tel: (852) 2898 2378
Fax: (852) 2898 1733
E-mail: kitty@swc.hk

2. or approved equivalent.

2.02 MATERIAL

- A. Provide and install vapour barrier, thermal insulations listed below or equivalent, and as shown on the Drawings, with all required accessory material necessary for a proper, complete installation meeting the requirements of this Section. (This indicates a partial lists of principal items only).
- B. Refer to the Contract Documents for items not specifically scheduled.
- C. Fire Stopping for Intumescent and Fire Stop Sealants: Refer to Section 6.4.
- D. "DP XPS X Series - grade 150" polystyrene insulation board

Insulation board shall be "DP XPS X Series - grade 150" self-extinguishing extruded type polystyrene board in 50mm thick for flat roof and podium deck

Other products submitted by the contractor may be considered for incorporation subject to compliance with the performance and properties, and approval by the Architect. The proprietary insulation board shall meet the following minimum performance:

Physical Properties	X150
Density (GB/T 10801.2-2002)	Min. 48.7kg/m ³
Thermal Coefficient, W / m x k (GB/T 10801.2-2002)	Max. 0.030
Compressive Strength, KPa (GB/T 10801.2-2002)	Min. 157
Water Permeability, ng/(m x s x Pa) (GB/T 10801.2-2002)	Max. 1.6
Water Absorption (96hours water immersion, %) (GB/T 10801.2-2002)	Max. 0.8
Dimension Stability, % (GB/T 10801.2-2002)	Length : Max. 0.5 Width : Max. 1.0 Thickness : Max. 1.5
Temperature Range	-50° to 75°C
Fire Resistance Value (GB 20286-23006)	B1 (self-extinguishing grade)
Heat Development Value, kW/m ² (GB 20286-23006)	Max. 237
Vertical Average Fire Spread Time, s (GB 20286-23006)	Max. 12
Vertical Average Fire Spread Height, mm (GB 20286-23006)	Max. 50
Smoke Density Rate, SDR (GB 20286-23006)	Max. 6
Thickness	50mm
Length (mm)	1200, 2400
Width (mm)	600
Lapping	Butt Joint

E. Thermobreak Insulation for Ceiling Soffit and under Raised Floor System

Physical Properties

Thickness:	25mm
Sheet Size:	1200 x 600mm
Density:	25 kg/m ³ (foam core only)
Thermal Conductivity: (ASTM C518)	0.032 W/mK (@ 23°C mean temp.)
Water Vapour Permeability: (ASTM E96)	2.3 x 10 ⁻¹⁵ kg/Pa.s.m
Water Vapour Permeance: 12mm thickness	0.000195 µg/N.s
Water absorption by volume: (ASTM C1763, Procedure B, 24h)	<0.2% v/v
Resistance to fungi: (ASTM G21)	Zero Growth
Ozone Resistance:	Excellent
UV Resistance:	Excellent
Noise Reduction Coefficient: (AS 1045)	0.20 (12mm foam thickness) 0.30 (25mm foam thickness)
Operating Temperature Range:	-80 °C ~ +100 °C (no adhesive)
GreenStar Rating: (ASTM D5116)	Low VOC Emitting
Physical Property Requirements: (ASTM C1427)	COMPLIES (Type II - Sheet)
REACH Directives: (1907/2006/EC)	COMPLIES

2.03 ANCILLARY MATERIALS

A. Vapour Barrier & Accessories :

1. All vapour barrier shall have a minimum vapour resistance of 15 MNs/g and non-combustible. because of the risk of interstitial condensation in cellular materials, additional vapour-sealing skins may be required.

2. Fire retardant vapour barrier: 5-ply polyvinyl chloride film and nylon yarn laminate reinforced with non-woven grid with a PPT tear strength of at least 156N (test method ASTM D-2582). Vapour resistance shall be as recommended by consulting Mechanical Engineer.
 3. Vapour Barrier Reinforced Tapes : 50 mm width.
- B. Wire Mesh: 0.55 mm diameter stainless steel, mesh at exterior conditions. Galvanised steel mesh for interior conditions.
 - C. Adhesive : Waterproof type, acceptable to manufacturer of insulation
 - D. Impaling Clips: Prefabricated galvanised steel clips, 1.6 mm thickness.
 - E. Firestop, barrier seals and fire-resistant sealants: Refer to Section 6.4.

PART 3 EXECUTION

3.01 PREPARATION

- A. Verify adjacent materials are dry and ready to receive installation.
- B. Verify mechanical, electrical, plumbing and other services within walls or other assemblies have been installed, tested and inspected.
- C. Verify insulation, insulation board, and vapour barrier are unbroken, and free of damage.
- D. Light switches, electrical outlets, junction boxes and processed panels within acoustically rated wall and other assemblies are not installed back-to-back, and that they are enclosed with a non-combustible, acoustic absorption and damping material.
- E. Preparation for Thermobreak Raised Floor Insulation:
 - 1. Use 50mm or greater width of recommended or specified reinforced aluminium foil tape.
 - 2. Fix pedestals as specified by the manufacturer (usually 600 mm apart).
 - 3. Install trunking supports before the insulation is applied. Do not at this stage install the trunking. The supports are fixed to the slab, therefore need to be installed before the insulation is applied.

3.02 INSTALLATION

- A. Install insulation in accordance with manufacturer's instruction, without gaps or voids.
- B. Trim insulation neatly to fit spaces. Use batts free of damage. Fit insulation tight in spaces and tight to exterior side of mechanical, plumbing, and electrical services within the plane of insulation.
- C. Where required, hold batt insulation in position with wire mesh. Secure in such a manner as to prevent movement or dislodgement of insulation, with no voids or gaps between batts.
- D. Install insulation with factory applied vapour barrier facing warm or conditioned side of building spaces. Lap ends and side flanges of vapour barrier and secure in place at 200 mm on centres. Tape seal butt ends and lapped side flanges of vapour barrier. Tape tears or cuts in vapour barrier. Ensure continuity of vapour barrier at all locations.
- E. Do not install insulation on top of, or within 75 mm of, recessed ceiling-mounted equipment or other heat-generating devices in order to prevent fixture, equipment, or device over heating.
- F. Insulation located on concrete or masonry walls shall be impaled on stick pins and then affix locking washer on to the pin to hold the insulation in place.
- G. Coordinate and refer to Work of Section 6.5 for sealing sound-isolating partition perimeters, cutouts and penetrations with Acoustical Sealant as specified by Fibre Cement Board Manufacturer at non-fire rated and fire-rated partitions.
- H. Installation for Thermobreak Raised Floor Insulation:
 - 1. Thermobreak Sheets (usually supplied in 1200mm x 600mm dimensions for raised floor applications) should be slit each corner at half way in order to fit the sheets between pedestals (Figure 8.1). The slit should be long enough to

accommodate the circumference of the pedestal. The flexibility of the foam allows a tight grip on the pedestal.

2. Apply contact adhesive to some parts of the insulation or concrete slab, including along the edges. Sekisui Foam International recommends a rubber/neoprene based contact adhesive. Follow instructions provided by the glue supplier.
3. Lower the insulation sheets so that the slits fit on the pedestals (Figure 8.2).
4. Seal all joints with aluminium foil tape as per Taping Instructions. Take care when walking on the insulation not to damage the foil facing.
5. Continue the above until the area is completely insulated.

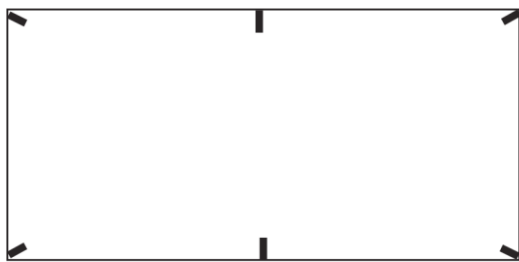


Figure 8.1. Slitting of Thermobreak Sheet.

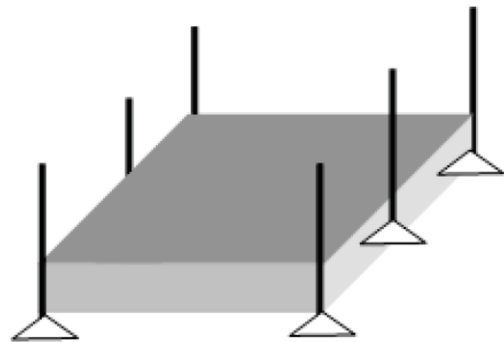


Figure 8.2. Thermobreak Sheet placed between pedestals.

3.03 PROTECTION

- A. Ensure that all vapour barriers are protected from damage during construction. Repair or replace all damaged vapour barrier before covering with other finish materials.
- B. Protect all insulation from damage from wetting, loading or abuse until they are under protective covering material or finish.

3.04 CLEANING

- A. Remove all debris after completion of the Work of this Section.

END OF SECTION